Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-08-C01-BD1

Brain dump 1 · foundational retrieval: Build the Virchow triad and the two thrombus types

A competent response contains

· Virchow triad: endothelial injury or dysfunction, abnormal flow (stasis or turbulence), and hypercoagulability; each limb can act alone but most thrombi need two or more.

· Endothelial injury exposes collagen and von Willebrand factor for platelet adhesion and tissue factor to start coagulation, and activated endothelium loses its anticoagulant surface; settings include ruptured atherosclerotic plaque, endocardium over a myocardial infarct, vasculitis, valve injury, hypertension, and smoking.

· Abnormal flow disrupts laminar flow so platelets reach the wall, stops activated factors being washed away and diluted, delays inflow of inhibitors, and turbulence itself injures endothelium; settings include immobility, atrial fibrillation, ventricular aneurysm, dilated chambers, aortic aneurysm, hyperviscosity, and sickle cell disease.

· Hypercoagulability is more procoagulant activity or less anticoagulant or fibrinolytic activity in the blood itself.

· Endothelial injury dominates in the heart and arteries; stasis dominates in veins.

· Arterial thrombus: pale gray, platelet rich with fibrin binding the platelets, firmly attached, usually starting on a ruptured plaque in coronary, cerebral, or femoral arteries, and growing back toward the heart from its point of attachment.

· Venous thrombus: soft red (red or stasis thrombus) because coagulation in nearly stagnant blood traps many red cells, grows in the direction of flow toward the heart, and can have a long loosely attached tail that breaks off.

· Most venous thrombi form in the deep veins of the leg, and those above the knee (popliteal, femoral, iliac) are the main source of pulmonary emboli.

· Mural thrombi form on the wall of a heart chamber or the aorta; vegetations are thrombi on valves.

· Normal endothelium keeps blood fluid by expressing thrombomodulin and heparan sulfate, releasing nitric oxide and prostacyclin, and making tissue plasminogen activator.

Distinctions that matter

· Composition follows flow rate, not vessel name: fast flow builds a platelet rich pale thrombus, slow flow builds a red cell rich soft one.

· Arterial thrombi grow back toward the heart against flow from their attachment, while venous thrombi grow toward the heart with flow; both end up pointing centrally for different reasons.

Common misconceptions

· Treating the three limbs as alternatives one of which must be chosen. Most thrombi arise where two or three act together, which is why a first thrombosis often needs an inherited trait plus a trigger.

· Saying stasis matters only because blood sits still. It also prevents dilution and washout of activated factors and delays the arrival of inhibitors.

Where it is taught: Section C01, opening paragraph, the Virchow triad table, and the composition paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C01-BD2

Brain dump 2 · integration and application: Tell a thrombus formed in life from one formed after death

A competent response contains

· The femoral vein mass is a postmortem clot: not attached to the wall, gelatinous, dark red where red cells settled with a yellow chicken fat layer above, and with no lines of Zahn.

· The left atrial mass is a thrombus formed during life: attached to the wall and showing lines of Zahn.

· Lines of Zahn are alternating pale bands of platelets and fibrin and dark bands of red cells, laid down in layers only while blood is moving, so they identify a thrombus formed before death.

· They are most obvious in the heart and aorta and fainter in veins.

· In still blood after death, cells sediment by gravity instead of layering, which gives the dark red and yellow division rather than banding.

· The thrombus that embolized formed in life; a postmortem clot cannot have embolized, so the attached layered mass is the plausible source.

· Fates of a thrombus: propagation (more platelets and fibrin, occluding the vessel or making a longer less stable thrombus), embolization (part or all breaks off and blocks a distant vessel), dissolution (plasmin degrades fibrin in a recent thrombus, while older cross-linked thrombi resist lysis), organization (endothelial cells, smooth muscle cells, and fibroblasts grow in, making granulation tissue then scar), and recanalization (new capillary channels form through the organized thrombus and partly restore flow).

· A fibrous thickening of the vein wall years later is an organized thrombus incorporated into the wall.

· Thrombi can also become infected, giving a septic thrombus or a mycotic aneurysm.

Distinctions that matter

· Attachment and layering are the two features that date a mass; color alone does not, since a venous thrombus formed in life is also red.

· Dissolution depends on the age of the thrombus: recent fibrin is lysed by plasmin, cross-linked fibrin resists it.

Common misconceptions

· Calling a red mass in a vein a postmortem clot because it is red and soft. A venous thrombus formed in life is red and soft too, and attachment plus lines of Zahn separate them.

· Treating recanalization as full recovery of the vessel. The channels are new small capillaries through organized thrombus, so flow is only partly restored.

Where it is taught: Section C01, lines of Zahn paragraph, the fates table, and the stop and predict box of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C02-BD1

Brain dump 1 · foundational retrieval: The brakes on thrombin and the inherited thrombophilias

A competent response contains

· Thrombin converts fibrinogen to fibrin and activates platelets and factors V, VIII, and XI.

· Brake one: antithrombin inactivates thrombin and factor Xa and other serine protease factors, and heparan sulfate on the endothelium greatly speeds the reaction.

· Brake two: thrombin binding thrombomodulin on endothelium activates protein C, and activated protein C with protein S as cofactor cleaves and inactivates factors Va and VIIIa.

· Brake three: plasmin dissolves fibrin.

· Factor V Leiden: point mutation in factor V (arginine 506 to glutamine), autosomal dominant; factor Va resists cleavage by activated protein C, so thrombin generation continues. Most common inherited cause, heterozygous in about 2 to 15% of people of European descent, risk about 5-fold heterozygous and about 50-fold homozygous.

· Prothrombin G20210A: mutation in the 3' untranslated region of the prothrombin gene raises plasma prothrombin; second most common, about 1 to 2% of the population.

· Antithrombin deficiency: reduced or defective antithrombin, so thrombin and factor Xa are not inactivated; heparin works by enhancing antithrombin, so its effect is blunted.

· Protein C or S deficiency: factors Va and VIIIa are not inactivated; homozygous protein C deficiency causes neonatal purpura fulminans.

· All four mainly cause venous thrombosis.

· Suspect an inherited thrombophilia when thrombosis occurs before about age 50, recurs, appears in unusual sites such as cerebral sinuses or mesenteric or hepatic veins, or runs in the family.

· Mild hyperhomocysteinemia is a further inherited or acquired risk; homocysteine injures endothelium and promotes arterial and venous thrombosis.

Distinctions that matter

· Factor V Leiden changes the substrate so the brake cannot act, while protein C or S deficiency removes the brake itself; the end result on factor Va is the same.

· Antithrombin acts on thrombin and factor Xa, while activated protein C acts on factors Va and VIIIa, so the two brakes work at different points of the cascade.

Common misconceptions

· Saying factor V Leiden means too much factor V. The amount is normal; the mutated factor Va simply resists cleavage by activated protein C.

· Expecting these disorders to cause arterial events. They mainly cause venous thrombosis, while antiphospholipid syndrome and hyperhomocysteinemia cause both.

Where it is taught: Section C02, opening paragraph and the inherited hypercoagulable states table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C02-BD2

Brain dump 2 · integration and application: Three clotting puzzles explained from the mechanism

A competent response contains

· Case one is antiphospholipid syndrome: autoantibodies bind phospholipid-binding proteins, mainly beta2-glycoprotein I, and activate platelets, endothelium, and complement, causing arterial and venous thrombosis and recurrent pregnancy loss.

· The lupus anticoagulant prolongs the partial thromboplastin time in the test tube because the antibodies interfere with the phospholipid in the assay, while the patient clots in vivo; anticardiolipin antibodies can give a false-positive nontreponemal syphilis test.

· Antiphospholipid syndrome may be primary or secondary to systemic lupus erythematosus.

· Case two: warfarin blocks vitamin K dependent synthesis of factors II, VII, IX, and X and of proteins C and S. Protein C has a short half-life, so protein C and factor VII fall first, before factors II, IX, and X.

· In those first days a person with low protein C is transiently more hypercoagulable, and dermal microvessels thrombose, which is warfarin-induced skin necrosis.

· Case three: pregnancy and the postpartum period raise hepatic synthesis of fibrinogen and other factors (hypercoagulability), lower protein S (hypercoagulability), and compress pelvic veins with the uterus (stasis).

· Acquired states are far more common than inherited ones and most act through more than one limb of the triad, so a first thrombosis often appears only when an acquired trigger is added to an inherited trait.

· Heparin is less effective in antithrombin deficiency because heparin works by enhancing antithrombin, and there is less of it to enhance.

Distinctions that matter

· A prolonged clotting time in vitro and thrombosis in vivo are not a contradiction: the antibody interferes with the phospholipid in the assay while activating cells in the patient.

· Warfarin skin necrosis is about half-lives, not dose: the brake disappears before the procoagulant factors do.

Common misconceptions

· Reading a long partial thromboplastin time as a bleeding risk in this syndrome. The patient clots, and the long time is an artifact of the assay.

· Assuming an inherited thrombophilia should have caused thrombosis earlier. Most carriers need an acquired trigger such as pregnancy, estrogen, surgery, or immobility.

Where it is taught: Section C02, warfarin paragraph, antiphospholipid paragraph, and the acquired hypercoagulable states table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-BD1

Brain dump 1 · foundational retrieval: Where an embolus lodges, and the nonthrombotic emboli

A competent response contains

· An embolus travels until it reaches a vessel too narrow to pass, so its source predicts its destination.

· Material entering systemic veins passes through the right heart to the pulmonary arteries; material from the left heart, aorta, or large arteries lodges in systemic arteries.

· A venous embolus reaches the systemic circulation only through a right-to-left opening such as a patent foramen ovale or atrial septal defect, which is paradoxical embolism.

· The great majority of emboli are dislodged thrombi.

· About 80% of systemic thromboemboli come from intracardiac mural thrombi: about two-thirds of those form over a left ventricular infarct and about a quarter in a dilated fibrillating left atrium; the rest come from aortic aneurysms, ulcerated plaques, valve vegetations, and paradoxical emboli.

· About 75% lodge in the lower limbs and about 10% in the brain, with the intestines, kidneys, spleen, and upper limbs making up the rest; these organs are supplied by end arteries, so infarction is the usual result.

· Fat and marrow embolism follows long-bone or pelvic fracture, less often soft tissue trauma or burns: fat globules block microvessels, free fatty acids released from them injure endothelium over the next days, and platelets adhere to fat and are consumed; fat embolism syndrome appears 1 to 3 days after injury with hypoxemia, tachypnea, neurologic change, petechial rash, and thrombocytopenia.

· Amniotic fluid embolism occurs in labor and the immediate postpartum period: fluid enters maternal veins through tears in membranes or uterine veins, triggering an anaphylactoid reaction and activating coagulation, with sudden dyspnea, cyanosis, shock, seizures, then disseminated intravascular coagulation, and fetal squames, lanugo hair, vernix fat, and mucin in maternal pulmonary vessels.

· Air embolism follows chest wall injury, obstetric procedures, or opened venous lines: bubbles coalesce and obstruct flow, and usually more than about 100 mL is needed for symptoms.

· Nitrogen embolism is decompression sickness from rapid ascent by divers or rapid decompression in flight, with joint and muscle pain (the bends) and respiratory distress (the chokes); chronic caisson disease causes ischemic necrosis of the femoral head, tibia, and humerus.

· Atheroembolism follows manipulation of an atherosclerotic aorta at catheterization or surgery: cholesterol crystals and plaque debris lodge in small arteries, giving renal failure, livedo reticularis, digital ischemia, and cholesterol clefts in arterioles.

· Septic emboli from infected vegetations and tumor emboli carry their infection or tumor to the site where they lodge.

Distinctions that matter

· The destination is decided by which side of the circulation the embolus entered, which is why a leg vein thrombus reaches the brain only through a shunt.

· Fat embolism and fat embolism syndrome are not the same: fat reaches the lungs in most patients with long-bone fracture, while the syndrome appears in a small minority after a delay.

Common misconceptions

· Expecting fat embolism syndrome at the moment of injury. The free fatty acid injury takes time, so the syndrome appears 1 to 3 days later.

· Assuming a small volume of air is fatal. Usually more than about 100 mL is needed for symptoms.

Where it is taught: Section C03, opening paragraph, systemic embolism paragraph, and the nonthrombotic emboli table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-BD2

Brain dump 2 · integration and application: Three pulmonary emboli of different size

A competent response contains

· Small peripheral embolus: usually clinically silent, and it organizes and is incorporated into the vessel wall.

· Medium embolus with left heart failure: hemorrhage would be the usual result because bronchial arteries keep supplying lung tissue, but left heart failure has impaired bronchial flow, so this patient infarcts, and the infarct is red because blood still enters from the bronchial circulation and congested veins.

· Saddle embolus at the bifurcation, obstructing more than about 60% of the pulmonary circulation: acute right heart failure (acute cor pulmonale), circulatory collapse, and sudden death.

· The embolized lung is ventilated but not perfused, so dead space rises; blood diverted to the remaining lung creates ventilation and perfusion mismatch and hypoxemia.

· Hypoxemia and stimulated receptors drive hyperventilation, so PaCO2 falls rather than rises.

· Repeated small emboli can cause pulmonary hypertension.

· More than 95% of pulmonary emboli arise from deep vein thrombi of the leg, most from veins above the knee, because thrombi in the larger proximal veins are big enough and loosely enough attached at the tail to break off and occlude a pulmonary artery branch.

Distinctions that matter

· Hemorrhage and infarction are different outcomes of the same sized embolus, and the state of the bronchial supply is what separates them.

· The lung in pulmonary embolism is hypoxemic with a low carbon dioxide, which is the opposite of what students expect from the word obstruction.

Common misconceptions

· Assuming every pulmonary embolus causes an infarct. The lung has two blood supplies, so most do not.

· Expecting carbon dioxide to rise because a vessel is blocked. Ventilation continues and is increased, so carbon dioxide falls.

Where it is taught: Section C03, paragraph on the effect of a pulmonary embolus, and the flow diagram caption of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-RT01

Retrieval exercise: Rebuild the route from a leg vein to a blocked pulmonary artery

A competent response contains

· Step 1: stasis, injury, or hypercoagulability in deep leg veins; stasis is the usual limb in a long flight or immobility, often with a hypercoagulable contribution.

· Step 2: a red thrombus propagates toward the heart.

· Step 3: a thrombus fragment detaches, usually the long tail, which is loosely attached because it was laid down in flowing blood away from the point of adhesion.

· Step 4: it passes through the right atrium and ventricle.

· Step 5: it lodges in a pulmonary artery branch.

· Step 6: perfusion is lost while ventilation of that lung continues, so ventilation is the one that is preserved and dead space rises.

· Chambers in order: deep leg vein, iliac vein, inferior vena cava, right atrium, right ventricle, pulmonary artery.

· A right-to-left shunt at the atrial level, a patent foramen ovale or atrial septal defect, would divert the fragment into the systemic arteries as a paradoxical embolus.

· More than 95% of pulmonary emboli arise from leg vein thrombi, most from veins above the knee.

Distinctions that matter

· The tail embolizes rather than the attached head, which is why venous thrombi growing with flow are the dangerous ones.

· Perfusion is lost and ventilation preserved, which is the definition of wasted ventilation rather than of shunt.

Common misconceptions

· Drawing the fragment traveling from the leg to the aorta directly. Without a right-to-left shunt it must stop in the lungs.

· Writing that ventilation stops in the embolized segment. Ventilation continues, which is why dead space rises.

Where it is taught: Section C03, the flow diagram from deep vein thrombosis to pulmonary embolism of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-BD1

Brain dump 1 · foundational retrieval: Draw an infarct and date it

A competent response contains

· An infarct is ischemic necrosis caused by loss of blood flow; nearly all follow arterial occlusion by thrombosis or embolism, with vasospasm, external compression, torsion, rupture, and venous outflow obstruction less common.

· Infarcts in solid organs are usually wedge shaped with the occluded vessel at the apex and the organ surface at the base, and a fibrinous exudate often covers the overlying serosa.

· The two questions that describe every infarct: how much blood it contains, white or red, and how the dead tissue breaks down, coagulative or liquefactive.

· Coagulative necrosis: ischemia denatures structural proteins and lytic enzymes alike, so cell outlines and tissue architecture persist for days while nuclei disappear.

· Time course: cell death is not visible by light microscopy for the first several hours; nuclear loss and eosinophilic cytoplasm appear over about 4 to 12 hours; neutrophils infiltrate the border within the first 1 to 2 days; macrophages and granulation tissue follow over the next week or two; the area is replaced by scar over weeks.

· The margin of a recent infarct is often hyperemic.

· The brain is the exception: little collagen framework and lipid-rich tissue digested by released lysosomal enzymes and cleared by microglia, so the infarct liquefies and leaves a fluid-filled cavity lined by reactive astrocytes rather than a scar.

· Color and necrosis pattern are independent: a brain infarct can be red and liquefactive, and a reperfused myocardial infarct red and coagulative.

· Septic infarcts, caused by infected emboli, become abscesses.

Distinctions that matter

· The necrosis pattern depends on the tissue, not on the color of the infarct.

· Neutrophils at the border and macrophages later are the clock that dates an infarct; the first several hours look normal under the microscope even though the cells are dead.

Common misconceptions

· Expecting a dead area to look dead immediately. Light microscopy shows nothing for several hours.

· Pairing white with coagulative and red with liquefactive. The two questions are answered separately.

Where it is taught: Section C04, opening paragraph and the paragraphs on necrosis pattern and time course of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-BD2

Brain dump 2 · integration and application: Same occlusion, four different outcomes

A competent response contains

· Patient one, renal artery branch: infarct forms, white, coagulative, because the kidney is a dense organ with end-arterial supply and no alternative inflow; weeks later it is a pale depressed scar.

· Patient two, pulmonary artery branch with a normal heart: usually hemorrhage without infarction, because bronchial arteries keep supplying the tissue; if infarction did occur it would be red, and the lung has a dual supply.

· Patient three, cerebral artery branch with lysis of the embolus and return of flow: infarct forms, red because blood enters damaged vessels when flow is restored, liquefactive because of the brain's lipid-rich tissue and scant collagen, and weeks later it is a fluid-filled cavity lined by reactive astrocytes.

· Patient four, slowly narrowed coronary artery: slow narrowing allowed collateral vessels to enlarge, so a small or no infarct forms; if one does form in the heart it is coagulative and white unless the artery is reperfused, in which case it is red, and it heals as scar.

· Factors deciding whether and how large an infarct forms: anatomy of the blood supply (dual supply in lung, liver, hand and forearm, or rich anastomoses protect, while end arteries in kidney and spleen do not), rate of occlusion (slow narrowing lets collaterals enlarge), vulnerability to hypoxia (neurons die after about 3 to 4 minutes, myocytes after about 20 to 30 minutes, fibroblasts survive many hours), and oxygen content of the blood (anemia, hypoxemia, or heart failure enlarge an infarct and can cause one after a partial occlusion).

· A hemoglobin of 7 g/dL would enlarge the infarct in patient four, because the slowly narrowed coronary artery leaves myocardium dependent on marginal flow, and reduced oxygen content converts a partial occlusion into a damaging one.

· These factors explain why the same occlusion causes no damage in one person and a large infarct in another.

Distinctions that matter

· Reperfusion changes the color of an infarct without changing its necrosis pattern.

· Rate of occlusion and anatomy of supply are separate protections: collaterals must be given time to enlarge, while a dual supply is there from the start.

Common misconceptions

· Treating low oxygen content as a cause of infarction by itself. It enlarges infarcts and can make a partial occlusion damaging, which is not the same as occluding a vessel.

· Expecting a brain infarct to leave a scar like the heart or kidney. It leaves a cavity, because the tissue liquefies.

Where it is taught: Section C04, white against red table, necrosis pattern paragraph, factors table, and the stop and predict box of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-CP01

Comparison table: White against red infarcts

A competent response contains

· Occlusion: white follows arterial occlusion; red follows venous occlusion, or arterial occlusion where blood can still enter.

· Tissue: white in solid dense organs with end-arterial supply; red in loose tissue, dual-supply organs, previously congested tissue, and reperfused tissue.

· Why: in a white infarct there is no alternative inflow and dense tissue limits seepage of blood into the necrotic area; in a red infarct blood from a second supply, from backed-up venous drainage, or from restored flow enters damaged vessels and leaks into the dead area.

· Examples: white in heart, kidney, and spleen; red in lung (pulmonary and bronchial arteries), small intestine (rich anastomoses), testis or ovary after torsion, brain after lysis of an embolus, and heart after reperfusion.

· Necrosis pattern: coagulative in almost every organ including heart, kidney, spleen, lung, and gut; liquefactive in the brain, whichever color the infarct is.

· Weeks later: a coagulative infarct is replaced by scar, which is pale, firm, and depressed; a brain infarct leaves a fluid-filled cyst lined by reactive astrocytes.

· Infarcts caused by infected emboli become abscesses.

Distinctions that matter

· Color answers where the blood came from; necrosis pattern answers what the tissue is made of. The two questions are independent.

· A red infarct is not a bleeding problem; it is dead tissue into which blood has entered from a second route.

Common misconceptions

· Filling the red infarct column with the brain only. Lung, gut, torsion, and reperfusion all belong there.

· Writing liquefactive for the lung because the lung is soft. Pattern follows the tissue, and only the brain liquefies.

Where it is taught: Section C04, white against red infarcts table and the necrosis pattern paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C05-BD1

Brain dump 1 · foundational retrieval: Patterns of bleeding into tissue

A competent response contains

· Petechiae: 1 to 2 mm pinpoint hemorrhages in skin, mucosa, or serosa that do not blanch, from thrombocytopenia, platelet dysfunction, raised intravascular pressure, or loss of vessel wall support as in scurvy.

· Purpura: 3 mm or larger up to about 1 cm, from the causes of petechiae plus vasculitis, trauma, and vascular fragility; palpable purpura suggests small-vessel vasculitis.

· Ecchymosis: a larger flat subcutaneous hemorrhage, a bruise, usually over 1 to 2 cm, from trauma and worsened by any bleeding disorder.

· Hematoma: a collected mass of blood within tissue, raised or space-occupying, from trauma, factor deficiencies such as hemophilia, or a ruptured vessel or aneurysm.

· Bleeding into cavities is named for the site: hemothorax, hemopericardium, hemoperitoneum, hemarthrosis.

· Pinpoint bleeds come from capillaries and venules normally sealed by platelets, so petechiae and purpura point to a platelet or small-vessel problem.

· Large deep bleeds into muscles, joints, and soft tissue follow minor injury when the fibrin clot that should stabilize the platelet plug cannot form, so they point to a coagulation factor deficiency.

· Bruise colors in order: red-blue from hemoglobin, then blue-green as heme becomes biliverdin and then bilirubin, then golden-brown from hemosiderin.

· A large hematoma can raise serum unconjugated bilirubin as it is resorbed.

Distinctions that matter

· Size separates petechiae, purpura, and ecchymosis; a hematoma is defined by being a mass within tissue rather than by size alone.

· Platelets seal small vessels, while fibrin stabilizes the plug in larger injuries, which is why the two kinds of defect bleed in different patterns.

Common misconceptions

· Reading all purpura as a platelet problem. Palpable purpura points to small-vessel vasculitis.

· Expecting bruise color change to come from the blood drying or oxidizing. It follows macrophage breakdown of hemoglobin through biliverdin and bilirubin to hemosiderin.

Where it is taught: Section C05, opening paragraph, the hemorrhage into tissues table, and the bruise paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C05-BD2

Brain dump 2 · integration and application: Volume, rate, and site in acute blood loss

A competent response contains

· Blood volume is about 70 mL/kg, so about 4.9 L in a 70 kg man; 1.5 L is about 30% of his blood volume.

· Rapid loss of up to about 10 to 15% is tolerated with little change; above about 20% compensations begin to fail and hypovolemic shock develops; about 40% is immediately life threatening.

· His hematocrit is normal because he lost whole blood, cells and plasma in the same proportion, so the concentration of red cells in what remains is unchanged even though red cell mass has fallen by about 30%.

· Two processes lower it over the next day: lower capillary hydrostatic pressure draws interstitial fluid into the vessels (transcapillary refill), and the kidney retains sodium and water through renin, angiotensin II, aldosterone, and ADH; this cell-free volume dilutes the remaining red cells.

· An early normal hematocrit therefore does not exclude severe acute blood loss.

· The baroreceptor reflex raises heart rate and constricts arterioles and veins, and the pulse pressure narrows.

· Site can matter more than volume: a small hemorrhage inside the rigid skull raises intracranial pressure and can cause herniation; a modest but rapid pericardial collection causes tamponade because the stiff sac cannot stretch quickly; retroperitoneal bleeding from a ruptured aortic aneurysm can hide liters of blood without external signs.

· Slow or recurrent external loss, as from a bleeding peptic ulcer or heavy menstruation, depletes iron and causes iron deficiency anemia, whereas blood shed into tissues or body cavities is broken down and its iron reused.

Distinctions that matter

· Volume lost and concentration measured are different things, which is why the hematocrit lags behind the hemorrhage.

· A rigid or non-compliant container turns a small volume into a dangerous one, so skull and pericardium behave unlike the retroperitoneum.

Common misconceptions

· Using a normal hematocrit to rule out major bleeding in the first hour.

· Expecting all blood loss to cause iron deficiency. Iron is reused when blood is shed internally and lost only when it leaves the body.

Where it is taught: Section C05, volume and rate paragraph, the worked example on hematocrit, and the final paragraph on site of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C06-BD1

Brain dump 1 · foundational retrieval: The mechanisms of edema and the fluid each produces

A competent response contains

· Edema is excess fluid in the interstitium.

· Normally capillaries filter slightly more fluid than they reabsorb and lymphatics return the excess; edema appears when filtration exceeds lymphatic capacity.

· Mechanisms: raised capillary hydrostatic pressure (heart failure, deep vein thrombosis or venous obstruction, portal hypertension with ascites, volume overload in renal failure) giving a transudate; lowered plasma oncotic pressure (nephrotic syndrome, cirrhosis, protein malnutrition or kwashiorkor, protein-losing enteropathy) giving a transudate; sodium and water retention (renal failure, renin, angiotensin, aldosterone activation in heart failure) giving a transudate; increased permeability (acute inflammation, burns, ARDS, anaphylaxis) giving an exudate; lymphatic obstruction (filariasis from Wuchereria bancrofti, tumor, lymph node dissection or radiation, congenital lymphatic hypoplasia) giving protein-rich lymph.

· Transudate: vessel wall intact, forces out of balance, water and small solutes cross while protein stays behind, so it is clear with low protein and low cell content and a specific gravity below about 1.012.

· Exudate: inflammatory mediators (histamine, bradykinin, leukotrienes) contract endothelial cells and open gaps, or the endothelium is directly injured, so protein, fibrinogen, and white cells leave, giving cloudy fluid with high protein and a specific gravity above about 1.020.

· Light criteria for pleural fluid mark an exudate: a fluid-to-serum protein ratio above 0.5, a fluid-to-serum LDH ratio above 0.6, or fluid LDH above two-thirds of the upper limit of normal for serum.

· Renal sodium and water retention adds to the hydrostatic and oncotic mechanisms by expanding plasma volume.

Distinctions that matter

· Transudate against exudate is a statement about the vessel wall, not about the volume of fluid or the organ involved.

· Lymphedema fluid is protein rich like an exudate but the wall is intact; the protein is there because lymph cannot be returned.

Common misconceptions

· Treating any high-protein fluid as inflammatory. Blocked lymphatics give protein-rich fluid without inflammation.

· Expecting one mechanism per patient. Heart failure raises hydrostatic pressure and adds renal retention at the same time.

Where it is taught: Section C06, opening paragraph, the mechanisms of edema table, and the transudate against exudate paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C06-BD2

Brain dump 2 · integration and application: Four swollen patients, four mechanisms

A competent response contains

· Nephrotic syndrome: albumin loss in the urine lowers plasma oncotic pressure, the fluid is a transudate with low protein, it pits, and it appears first in loose tissue such as the eyelids.

· Heart failure: raised venous and therefore capillary hydrostatic pressure, with renal sodium and water retention adding plasma volume; transudate, pits, and collects in the legs when standing and over the sacrum when lying.

· Axillary node dissection: lymphatic obstruction, so retained fluid is protein-rich lymph, the arm is affected, and it begins soft and pitting but becomes nonpitting.

· Lymphedema becomes hard because retained protein stimulates fibroblasts and adipose deposition, so skin and subcutaneous tissue thicken; over years this is elephantiasis, and blocked dermal lymphatics in breast cancer give the dimpled peau d'orange skin.

· Hypothyroid myxedema is nonpitting for a different reason: glycosaminoglycans bind water in the dermis, so the fluid is not free in the interstitium.

· Ordinary edema pits because the excess fluid is free in the interstitium: firm pressure pushes it aside and the dent fills slowly.

· It collects where hydrostatic pressure is highest, which is the legs when standing and the sacrum when lying.

Distinctions that matter

· Pitting depends on whether the fluid is free to move, which is why both fibrotic lymphedema and myxedema fail to pit for different reasons.

· Nephrotic edema starts in loose tissue such as the eyelids because the oncotic deficit is systemic, while hydrostatic edema starts where pressure is highest.

Common misconceptions

· Calling all nonpitting swelling lymphedema. Myxedema is nonpitting from bound glycosaminoglycans.

· Describing lymphedema as permanently soft. It starts soft and becomes hard as protein drives fibrosis.

Where it is taught: Section C06, mechanisms table and the final paragraph on pitting and lymphedema of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C06-RT01

Retrieval exercise: Rebuild the edema cycle in heart failure

A competent response contains

· Step 1: cardiac output falls.

· Step 2: renal perfusion falls and renin rises.

· Step 3: angiotensin II and aldosterone retain sodium and water; ADH contributes water retention.

· Step 4: plasma volume and venous pressure rise.

· Step 5: capillary hydrostatic pressure rises and edema forms; the changed Starling force is capillary hydrostatic pressure and the fluid is a transudate.

· Step 6: the failing heart is further overloaded, which lowers output again and closes the loop.

· Renal retention is meant to restore perfusion by raising filling, but in a failing ventricle the extra volume raises venous pressure and worsens both edema and the load on the heart.

· Lowering venous pressure interrupts the loop at step four, before capillary hydrostatic pressure rises.

Distinctions that matter

· The same volume expansion that would help a hypovolemic patient harms a failing ventricle, because the ventricle cannot translate added filling into output.

· Hydrostatic edema in heart failure is driven by two mechanisms at once, the raised venous pressure and the renal retention that feeds it.

Common misconceptions

· Reading the renal response as a malfunction. It is a normal response to a low perfusion signal, which is why it fires in a patient who is not volume depleted.

· Calling this fluid an exudate because it is caused by disease. The vessel wall is intact, so it is a transudate.

Where it is taught: Section C06, the edema in heart failure cycle of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-BD1

Brain dump 1 · foundational retrieval: Build the hemodynamic table of shock

A competent response contains

· Shock is widespread tissue hypoperfusion in which oxygen delivery falls below metabolic demand, from low cardiac output or low effective circulating volume.

· Mean arterial pressure equals cardiac output times systemic vascular resistance, and cardiac output depends on venous return, so perfusion fails when the volume, the pump, the path through the heart, or vascular tone fails.

· The three measurements are cardiac output, systemic vascular resistance, and pulmonary capillary wedge pressure, which approximates left atrial pressure and so left ventricular filling pressure.

· Hypovolemic: loss of blood or fluid (hemorrhage, burns, vomiting, diarrhea); output decreased, resistance increased as a reflex, wedge decreased, skin cold and clammy.

· Cardiogenic: pump failure (myocardial infarction, myocarditis, arrhythmia, acute valve rupture); output decreased, resistance increased as a reflex, wedge increased, skin cold and clammy.

· Obstructive: block to filling or outflow (tamponade, massive pulmonary embolism, tension pneumothorax); output decreased, resistance increased as a reflex, wedge increased in tamponade with equal diastolic pressures and normal or low in pulmonary embolism, skin cold and clammy.

· Septic (distributive): vasodilation and capillary leak from the host response to infection; output increased or normal early and may fall late, resistance decreased, wedge decreased or normal, skin warm and flushed early.

· Anaphylactic (distributive): mast cell and basophil release of histamine and other mediators, usually IgE mediated; output usually decreased by vasodilation and leak and may be normal, resistance decreased, wedge decreased, skin warm and flushed with urticaria and angioedema.

· Neurogenic (distributive): loss of sympathetic tone after spinal cord injury usually above about T6; output decreased, resistance decreased, wedge decreased, skin warm and dry with bradycardia.

· Low-output shock raises resistance as a reflex; distributive shock lowers resistance as the primary defect.

· Normal values: cardiac output about 4 to 8 L/min, systemic vascular resistance about 800 to 1,200 dyn.s/cm5, pulmonary capillary wedge pressure about 6 to 12 mm Hg.

· Central venous pressure follows the wedge pressure except in right-sided obstruction such as pulmonary embolism, where it rises while the wedge stays normal or low.

Distinctions that matter

· High resistance in low-output shock is a response, while low resistance in distributive shock is the disease; naming which is primary is the whole point of the table.

· Neurogenic shock is distributive but shares a low cardiac output with the low-output types, and bradycardia with warm dry skin is what marks it out.

Common misconceptions

· Grouping all three distributive types as high output. Anaphylactic and neurogenic shock usually have a low output, and only early septic shock is reliably hyperdynamic.

· Expecting the wedge pressure to be high in all obstructive shock. It is high in tamponade and normal or low in massive pulmonary embolism.

Where it is taught: Section C07, opening paragraph and the hemodynamic profiles table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-BD2

Brain dump 2 · integration and application: Read four hemodynamic profiles

A competent response contains

· Patient A: cardiogenic shock. Output 2.5 L/min is low and is the primary change, resistance 1,800 is a reflex, and a wedge of 26 mm Hg means the left ventricle is full but not emptying, which also causes pulmonary edema.

· Patient B: hypovolemic shock. Output 2.8 L/min is low, resistance 1,700 is a reflex, and a wedge of 4 mm Hg shows the ventricle is empty, so reduced venous return is the primary problem.

· Patient C: early septic shock. Resistance of 450 is the primary defect, and an output of 8.5 L/min has risen because low resistance and reflex tachycardia allow it; the wedge of 8 mm Hg is normal.

· Patient D: obstructive shock from massive pulmonary embolism. Output is low with reflex high resistance, the central venous pressure of 20 mm Hg is raised because the right ventricle cannot push blood through the lungs, and the wedge of 7 mm Hg is normal or low because the left ventricle is underfilled.

· Central venous pressure and wedge pressure diverge when the obstruction sits between them, which happens in massive pulmonary embolism; tension pneumothorax is the other cause of obstructive shock with raised right-sided pressure.

· In tamponade, fluid around the heart limits filling of all chambers, so diastolic pressures in the atria, ventricles, and the wedge rise and equalize.

· Patient C has a normal or high mixed venous oxygen saturation; tissues remain hypoxic because flow is maldistributed and mitochondrial oxygen use is impaired, unlike the low saturation of low-output shock.

Distinctions that matter

· A low output with a high wedge separates cardiogenic from hypovolemic shock, since both have low output and high resistance.

· Mixed venous saturation separates maldistributed flow from inadequate flow: it is high when blood passes tissues without giving up oxygen and low when there is not enough flow.

Common misconceptions

· Reading a high resistance as vasoconstrictor disease. In these patients it is the baroreceptor reflex doing its job.

· Assuming a high cardiac output rules out shock. Early septic shock is hyperdynamic and the tissues are still hypoxic.

Where it is taught: Section C07, hemodynamic profiles table, the paragraphs on low-output and distributive shock, and the worked example of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-CP01

Comparison table: Cardiogenic against hypovolemic against septic shock

A competent response contains

· Primary problem: cardiogenic is pump failure from myocardial infarction, myocarditis, arrhythmia, or acute valve rupture; hypovolemic is loss of blood or fluid from hemorrhage, burns, vomiting, or diarrhea; septic is vasodilation and capillary leak from the host response to infection.

· Cardiac output: decreased and primary in cardiogenic; decreased in hypovolemic as a consequence of reduced venous return; increased or normal early in septic shock as a response to low resistance, and it may fall late.

· Systemic vascular resistance: increased as a reflex in cardiogenic and hypovolemic shock; decreased as the primary defect in septic shock.

· Wedge pressure: increased in cardiogenic shock, showing a full ventricle that cannot empty; decreased in hypovolemic shock, showing an empty ventricle; decreased or normal in septic shock.

· Skin: cold and clammy in cardiogenic and hypovolemic shock from intense arteriolar constriction; warm and flushed early in septic shock from vasodilation.

· Mixed venous oxygen saturation: low in cardiogenic and hypovolemic shock because tissues extract more from a reduced flow; normal or high in septic shock because flow is maldistributed and mitochondrial oxygen use is impaired.

· Why tissues are underperfused: in cardiogenic shock the pump cannot deliver, in hypovolemic shock there is too little volume to deliver, and in septic shock there is flow but it is distributed away from nutrient capillaries and cannot be used.

Distinctions that matter

· Cardiogenic and hypovolemic shock look identical except at the wedge pressure, which is why that one measurement decides between them.

· Septic shock is the one column where the cardiac output can be normal or high while the patient is in shock.

Common misconceptions

· Expecting warm skin to mean a patient is not in shock. Vasodilation is the defect in septic shock.

· Writing a low mixed venous saturation for septic shock by analogy with the other two. Maldistributed flow and impaired oxygen use raise it.

Where it is taught: Section C07, hemodynamic profiles table and the paragraphs on low-output and distributive shock of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-PR01

Prediction exercise: Predict the profile after a spinal cord injury at T4

A competent response contains

· Sympathetic outflow to arterioles and veins below the injury is lost, because the cord injury interrupts the sympathetic pathways leaving the cord.

· Systemic vascular resistance falls, and the fall is the primary defect rather than a reflex, which makes this distributive shock.

· Venous return falls because venous capacitance rises and blood pools, so the wedge pressure falls.

· Cardiac output falls, because the reduced venous return lowers filling and stroke volume.

· Heart rate falls: cut sympathetic pathways cannot raise it and unopposed vagal tone causes bradycardia, unlike every other low-output shock, where the baroreceptor reflex produces tachycardia.

· The skin is warm and dry, because arteriolar constriction and sympathetic sweating are both lost; in hypovolemic shock intense reflex constriction makes the skin cold and clammy.

· Neurogenic shock follows cord injury usually above about T6.

Distinctions that matter

· Neurogenic shock has a low output and a low resistance together, which separates it from both hypovolemic and septic profiles.

· Bradycardia with hypotension is the single sign that points away from every reflex-intact form of shock.

Common misconceptions

· Expecting tachycardia because the blood pressure is low. The reflex pathway to the heart is interrupted.

· Calling the skin warm and flushed as in sepsis. It is warm and dry, since sympathetic sweating is lost as well.

Where it is taught: Section C07, hemodynamic profiles table and the distributive shock paragraph of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-BD1

Brain dump 1 · foundational retrieval: The three stages of shock and the organs that fail

A competent response contains

· Nonprogressive (compensated) stage: baroreceptor reflexes, catecholamines, the renin, angiotensin, and aldosterone system, and ADH raise heart rate, constrict arterioles in skin, gut, and kidney, and retain salt and water; autoregulation keeps coronary and cerebral flow nearly normal, so there is tachycardia, cool pale skin except in early sepsis, and reduced urine output, with a blood pressure that may be near normal.

· Progressive stage: perfusion is inadequate despite the reflexes; cells shift to anaerobic glycolysis and lactate accumulates, giving a high anion gap metabolic acidosis with serum lactate above the normal of about 2 mmol/L; acidosis blunts arteriolar responses to sympathetic stimulation while venules stay constricted, so blood pools and leaks, and it depresses contractility; patients become confused, oliguric, and hypotensive.

· Irreversible stage: restoring hemodynamics no longer saves the patient; ATP is depleted, lysosomal enzymes leak from injured cells, the myocardium fails, and ischemic bowel lets bacteria and endotoxin enter the blood, adding sepsis to the original insult.

· Kidney: acute tubular necrosis, because the proximal straight tubule and thick ascending limb in the outer medulla have high energy demand and little oxygen; oliguria, rising creatinine, muddy brown granular casts.

· Lung: ARDS or diffuse alveolar damage from neutrophil and cytokine injury to endothelium and type I pneumocytes; protein-rich edema, hyaline membranes, hypoxemia with a normal left atrial pressure.

· Blood: disseminated intravascular coagulation, where microthrombi consume platelets and factors, giving low platelets and fibrinogen, long prothrombin and partial thromboplastin times, high D-dimer, and schistocytes.

· Liver: ischemic hepatitis or shock liver, with centrilobular zone 3 necrosis and very high aminotransferases.

· Gut, brain, and heart: mucosal necrosis, encephalopathy, and subendocardial necrosis, since areas perfused last fail first, and gut injury allows bacterial translocation.

· Adrenal: hemorrhagic necrosis, with bilateral adrenal hemorrhage in meningococcal sepsis, the Waterhouse-Friderichsen syndrome.

· The rule: each organ fails where its blood supply is weakest relative to its metabolic demand.

Distinctions that matter

· The compensated stage can have a near normal blood pressure, so urine output, skin, and heart rate are what reveal it.

· Acidosis in the progressive stage is both a consequence of hypoperfusion and a cause of further hypoperfusion, which is what makes the stage progressive.

Common misconceptions

· Treating hypotension as the first sign of shock. Reflexes hold the pressure up through the whole compensated stage.

· Expecting the irreversible stage to be defined by a particular pressure. It is defined by the patient no longer recovering when hemodynamics are restored.

Where it is taught: Section C08, the three stage paragraphs and the multiorgan dysfunction table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-BD2

Brain dump 2 · integration and application: From one gram negative bacterium to failure in five organs

A competent response contains

· Septic shock is caused by the host response, not by the bacteria directly.

· The molecule is lipopolysaccharide, endotoxin of gram-negative bacteria, and lipid A is its toxic portion.

· It binds LPS-binding protein, CD14, and TLR4 on macrophages, neutrophils, and endothelium; gram-positive wall components signal through TLR2, and superantigens activate T cells nonspecifically.

· NF-kappa B is activated and drives release of TNF and IL-1, which induce IL-6 and chemokines, while microbial products activate complement to generate C3a and C5a.

· Inducible nitric oxide synthase in endothelium and smooth muscle makes large amounts of nitric oxide, which relaxes vascular smooth muscle through cGMP and lowers systemic vascular resistance; this is the warm flushed skin and the hypotension.

· TNF and IL-1 loosen endothelial junctions, which gives the capillary leak and a hypovolemic component, and they depress myocardial contraction.

· Cytokines push coagulation toward clotting: they induce tissue factor, reduce thrombomodulin so less protein C is activated, and raise plasminogen activator inhibitor-1, suppressing fibrinolysis; microthrombi and consumption of platelets and factors produce disseminated intravascular coagulation, which is why the tests show clotting and bleeding at the same time, with a falling platelet count and a rising D-dimer.

· The rising creatinine is acute tubular necrosis, which appears in the outer medulla because the proximal straight tubule and thick ascending limb there have high energy demand and little oxygen.

· The hypoxemia is ARDS: neutrophil and cytokine injury to endothelium and type I pneumocytes gives protein-rich edema and hyaline membranes, so the left atrial pressure is normal and the edema is not from a failing left ventricle.

· Her shock would keep worsening because the positive feedback cycle of the progressive stage is now self-sustaining: each fall in flow causes a further fall even if the original insult stops.

· In the irreversible stage, restoring hemodynamics no longer helps.

Distinctions that matter

· Low resistance, capillary leak, and microthrombi all come from the same cytokine-activated endothelium, which is why one infection produces vasodilation, hypovolemia, and clotting together.

· ARDS edema and cardiogenic pulmonary edema are separated by the left atrial pressure: normal in ARDS, raised in left ventricular failure.

Common misconceptions

· Blaming the organ damage on the bacteria spreading to each organ. The host cytokine response and hypoperfusion do the damage.

· Reading a long prothrombin time with a low platelet count as a primary bleeding disorder. Consumption by microthrombi is what lengthens the times.

Where it is taught: Section C08, septic shock mediators paragraph, the endotoxin flow diagram, the coagulation paragraph, and the multiorgan dysfunction table of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-RT01

Retrieval exercise: Rebuild the endotoxin sequence and the vicious cycle

A competent response contains

· Step 1: endotoxin binds CD14 and TLR4 on macrophages and endothelium.

· Step 2: NF-kappa B is activated and TNF and IL-1 are released.

· Step 3: inducible nitric oxide synthase makes nitric oxide, which relaxes vascular smooth muscle through cGMP.

· Step 4: endothelial leak develops, tissue factor goes up and protein C activation goes down, so microthrombi form in small vessels.

· Step 5: low systemic vascular resistance, hypovolemia, microthrombi, and myocardial depression.

· Step 6: hypoperfusion and multiorgan dysfunction.

· Vasodilation, leak, and clotting all arise from the same cytokine-activated endothelium.

· Progressive shock cycle: cardiac output and arterial pressure fall; tissue hypoxia with lactic acidosis; arterioles dilate and blood pools in capillaries; capillary leak and venous return falls; acidosis depresses the myocardium; cardiac output falls further.

· The two points where falling flow causes a further fall are the pooling and leak step, which lowers venous return, and the acidosis step, which depresses the myocardium.

· Once each fall in flow causes a further fall, shock worsens even if the original insult stops.

Distinctions that matter

· Nitric oxide lowers resistance while TNF and IL-1 open junctions and depress the myocardium, so the three hemodynamic problems have three separate mediators.

· Loss of fibrinolysis matters as much as the gain in tissue factor, which is why the microthrombi persist.

Common misconceptions

· Treating the cycle as needing continued infection. It is self-sustaining once started.

· Expecting the venules to dilate with the arterioles. They stay constricted, which is what traps blood in the capillary bed.

Where it is taught: Section C08, the endotoxin flow diagram and the positive feedback cycle of progressive shock of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C08-CA01

Clinical application: Hypotension and bleeding gums three days after a fracture

A competent response contains

· Recognize: septic shock from gram-negative infection, in the progressive stage, with acute tubular necrosis in the kidney, ARDS in the lung, and disseminated intravascular coagulation in the blood.

· Explain: lipopolysaccharide with lipid A as its toxic portion binds LPS-binding protein, CD14, and TLR4; NF-kappa B activation releases TNF and IL-1, which induce IL-6 and chemokines; inducible nitric oxide synthase makes nitric oxide, which relaxes vascular smooth muscle through cGMP and lowers resistance, giving warm flushed skin and hypotension, while TNF and IL-1 loosen endothelial junctions to produce the leak.

· Differentiate: her resistance is low as the primary defect with a normal or high output and a normal or low wedge pressure, and her skin is warm, whereas cardiogenic and hypovolemic shock both have a low output with reflex high resistance and cold clammy skin; the wedge pressure separates those two from each other, being high in cardiogenic and low in hypovolemic shock, and systemic vascular resistance is the measurement that marks hers out as distributive.

· Investigate: for acute tubular necrosis, creatinine and urine output and the urine sediment, where muddy brown granular casts appear; for ARDS, arterial blood gas showing hypoxemia with a chest radiograph showing bilateral infiltrates and a left atrial or wedge pressure that is normal; for disseminated intravascular coagulation, platelet count and fibrinogen falling, prothrombin and partial thromboplastin times lengthening, D-dimer rising, and schistocytes on the smear.

· Decide: disseminated intravascular coagulation explains both, since microthrombi form while platelets and clotting factors are consumed in making them, leaving nothing to seal her gums and puncture sites; at the progressive stage her shock can still respond to restoring the circulation, but the positive feedback cycle means it will keep worsening until it is interrupted.

· Justify: cytokines induce tissue factor, reduce thrombomodulin so less protein C is activated, and raise plasminogen activator inhibitor-1 to suppress fibrinolysis, so clotting is driven and clot removal is blocked; the irreversible stage is defined by hemodynamic restoration no longer helping, with ATP depletion, lysosomal enzyme leak, myocardial failure, and bacterial translocation from ischemic bowel, and she has not reached that point.

Distinctions that matter

· Thrombosis and bleeding in the same patient are one process at two stages: microthrombi first, then consumption of what made them.

· Progressive and irreversible stages are separated by whether restoring hemodynamics still helps, not by how abnormal the numbers are.

Common misconceptions

· Reading warm flushed skin with hypotension as a reason to look away from shock.

· Taking her bilateral infiltrates as cardiogenic pulmonary edema. In ARDS the left atrial pressure is normal and the edema is protein rich from injured endothelium.

Where it is taught: Section C08, stages paragraphs, mediators paragraph, coagulation paragraph, and the multiorgan dysfunction table, with the hemodynamic profiles table in C07 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C03-Q01

Board-style question: Sudden collapse after a flight

Correct answer: A. Hyperventilation driven by hypoxemia and stimulated receptors, with ventilation continuing in lung that is no longer perfused

Why it is right: Hypoxemia and stimulated receptors increase total ventilation, and carbon dioxide is far more diffusible and more responsive to ventilation than oxygen, so the hyperventilation of the perfused lung lowers PaCO2 while the ventilated but unperfused segment adds dead space and does not clear carbon dioxide at all.

B is wrong: Reduced carbon dioxide production by hypoperfused tissues is far too small and too slow to account for an acute fall, and hypoperfused tissues in fact generate an acid load from anaerobic glycolysis.

C is wrong: The affected segments are still ventilated rather than collapsed, which is exactly why dead space rises, and loss of alveolar surface would raise carbon dioxide rather than lower it.

D is wrong: The disturbance here is a respiratory one driven by increased ventilation, not a metabolic alkalosis, and catecholamine release does not remove acid or add bicarbonate.

A competent response contains

· A saddle embolus obstructing more than about 60% of the pulmonary circulation causes acute right heart failure, circulatory collapse, and sudden death.

· Embolized lung is ventilated but not perfused, so dead space rises, and hypoxemia drives hyperventilation that lowers PaCO2.

Distinctions that matter

· Perfusion is lost while ventilation continues, so the gas pattern is a low oxygen with a low carbon dioxide rather than retention.

Common misconceptions

· Expecting carbon dioxide to rise whenever a pulmonary vessel is blocked.

Where it is taught: Section C03 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C07-Q02

Board-style question: A profile with a high filling pressure

Correct answer: C. Failure of the left ventricle to empty, with reflex arteriolar constriction

Why it is right: Pump failure lowers cardiac output, the baroreceptor reflex raises systemic vascular resistance and makes the skin cool and mottled, and blood backing up behind the ventricle that cannot empty raises the wedge pressure to 28 mm Hg and floods the lungs.

A is wrong: Loss of circulating volume also gives a low output with reflex constriction, but the ventricle would be underfilled and the wedge pressure low rather than 28 mm Hg.

B is wrong: Reduced sympathetic tone would lower systemic vascular resistance rather than raise it to 1,900, and it would give warm dry skin with bradycardia.

D is wrong: Cytokine-driven vasodilation with leak lowers resistance and leaves the wedge pressure normal or low, and the skin would be warm and flushed early.

A competent response contains

· A low cardiac output with a high systemic vascular resistance and a high wedge pressure is cardiogenic shock, here after a myocardial infarction.

· The high wedge pressure means the left ventricle is full but not emptying, which also explains the crackles from pulmonary edema.

Distinctions that matter

· The wedge pressure is the one measurement that separates cardiogenic from hypovolemic shock, since both have low output and reflex high resistance.

Common misconceptions

· Reading the high resistance as the primary problem when it is the baroreceptor reflex responding to a low output.

Where it is taught: Section C07 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C02-Q03

Board-style question: A long clotting time with thrombosis

Correct answer: B. Antibodies against beta2-glycoprotein I that activate platelets, endothelium, and complement

Why it is right: Antibodies to beta2-glycoprotein I activate platelets, endothelium, and complement, which produces arterial and venous thrombosis and recurrent pregnancy loss, while binding the phospholipid in the laboratory assay prolongs the partial thromboplastin time and anticardiolipin antibodies cross-react in nontreponemal syphilis testing.

A is wrong: Factor V Leiden causes activated protein C resistance and venous thrombosis, but it does not prolong the partial thromboplastin time and does not cause a false-positive syphilis screen.

C is wrong: Antithrombin deficiency leaves thrombin and factor Xa uninactivated and blunts the effect of heparin, but it does not produce an inhibitor pattern on mixing or a positive syphilis screen.

D is wrong: Prothrombin G20210A raises plasma prothrombin and causes venous thrombosis, with no effect on the partial thromboplastin time and no cross-reacting antibody.

A competent response contains

· Venous thrombosis with recurrent pregnancy loss, a prolonged partial thromboplastin time that does not fully correct on mixing, and a false-positive nontreponemal syphilis test is antiphospholipid syndrome.

· Antibodies bind phospholipid-binding proteins, mainly beta2-glycoprotein I, and activate platelets, endothelium, and complement, causing arterial and venous thrombosis.

Distinctions that matter

· An inhibitor in the plasma does not correct on mixing with normal plasma, whereas a factor deficiency does, which is why the long time here points to an antibody rather than a missing factor.

Common misconceptions

· Taking a long clotting time as evidence of bleeding risk. The antibody interferes with the phospholipid in the assay while the patient clots in vivo.

Where it is taught: Section C02 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.

NPLEX-P1-CARD-18-08-C04-Q04

Board-style question: A cavity where an infarct had been

Correct answer: D. Brain tissue has little collagen framework and is lipid rich, so lysosomal enzymes liquefy it and microglia clear it away

Why it is right: In almost every organ ischemia denatures structural proteins and lytic enzymes alike, so architecture persists and the area is replaced by scar, while the brain's scant collagen and lipid-rich tissue allow lysosomal enzymes to liquefy it and microglia to clear the debris, leaving a cyst lined by reactive astrocytes.

A is wrong: Color records where blood came from and does not decide healing, and a white infarct in a solid organ scars while a brain infarct cavitates whether it is red or white.

B is wrong: Reperfusion after lysis of an embolus makes a brain infarct red, but the liquefaction and the cavity follow from the tissue itself, so an unreperfused brain infarct also cavitates.

C is wrong: Differing tolerance of hypoxia decides how quickly and how widely cells die, not what the dead tissue becomes afterwards.

A competent response contains

· The necrosis pattern depends on the tissue rather than on the color of the infarct.

· The brain has little collagen framework and lipid-rich tissue that released lysosomal enzymes digest and microglia clear, so the infarct liquefies and leaves a fluid-filled cavity lined by reactive astrocytes, while coagulative necrosis elsewhere is replaced by scar.

Distinctions that matter

· Color and necrosis pattern are separate questions, so a red brain infarct and a white kidney infarct still heal according to their tissue.

Common misconceptions

· Assuming blood in the dead tissue decides how an infarct heals.

Where it is taught: Section C04 of the CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock notes.